

# Hernan "Ernie" Maldonado

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## Summary

Data Engineer & ML Specialist with 10+ years of experience architecting large-scale pipelines (Hadoop, Spark, Hive, Oozie, SQL) and production ML models for fraud, compliance, and cybersecurity at major financial institutions. Proven track record processing hundreds of TBs of data, slashing processing times by 90%, and deploying production-grade NLP and gradient-boosted systems. Credentials include 3 graduate degrees, 4 advanced ML certifications, and an AWS Certified Cloud Practitioner badge; currently pursuing the AWS Certified Data Engineer – Associate certification. Targeting Data Engineering, ML Engineering, or Generative AI roles.

## Core Skills

Python • SQL • PySpark • Hadoop/Hive • XGBoost • Scikit-Learn • AWS • Docker • FastAPI

## Work Experience

### Internal Fraud Engineer

Sep 2024 – Present

*Citigroup, Tampa FL*

*Hybrid*

- Mine Hadoop databases (transactions, user behavior, historical cases) to surface fraud patterns and translate findings into **PySpark/Hive** detection rules targeting anomalous velocity, high-risk access, and suspicious vendors.
- Collaborate with analysis teams to simulate, tune, and deploy alert logic; manage alert volume by focusing on recurring MOs that mirror confirmed past crimes.

### Consumer Protection Big Data & ML Consultant

Oct 2023 – Dec 2023

*Bank of America, Tampa FL*

*Hybrid*

- Audited the entire fraud **Hadoop** ecosystem and produced a data-driven benchmark report comparing current practices against industry standards across performance, reliability, and data quality.

### Sanctions Machine Learning Engineer

May 2022 – Oct 2023

*Citigroup, Tampa FL*

*Hybrid*

- Automated 500K monthly sanctions alerts by building end-to-end NLP pipelines (Random Forest, Gradient Boosting, XGBoost) with 50+ engineered features to match flagged customers against sanctioned-persons lists.
- Cut manual review volume by 50% by discovering Citi's internal transactions were being double-screened across two independent systems.
- Authored Model Risk Management documentation ensuring compliance with OCC 2011-12 and SR 11-7; produced architecture study showing a Hadoop migration would reduce processing time by 90%.
- **Stack:** SQL, Pandas, Python, Scikit-Learn, PyTorch, XGBoost, FastAPI

### Data Scientist – Fraud Detection ML

May 2021 – Apr 2022

*Certegy Payments, Tampa FL*

*Hybrid*

- Doubled check-fraud detection rate at Walmart by developing Random Forest, Gradient Boosting, and Logistic Regression models; raised KS statistic from 0.05–0.3 to 0.6–0.8.
- Architected full ML pipeline: features built in Oracle SQL, models served via AWS API with real-time and batch scoring, routed to the correct model per transaction type.
- **Stack:** SAS Viya, Oracle SQL, Oracle OML, Python, PyTorch, Scikit-Learn, TensorFlow, AWS

### Sr. Data Scientist

Feb 2021 – Apr 2021

*Nielsen, Tampa FL*

*Remote*

- Maintained and extended Java codebase for national-scale TV viewership adjudication; developed new attribution features for programming and advertising data. **Stack:** Java, S3, Git

### Cybersecurity Big Data & ML Engineer

Apr 2018 – Feb 2021

*Citigroup, Tampa FL*

*Hybrid*

- Built from scratch a production anomaly-detection system monitoring 300K employees across 5TB of weekly data using peer-normalized behavioral metrics and k-means clustering to surface insider threats for the Cyber Security Fusion Center.
- Engineered ingestion pipelines for 12+ data sources into a 270TB Hadoop cluster (Sqoop, Flume, Python parsers, Hive, Oozie) and built a monitoring system to ensure all feed flows remained current.
- **Stack:** PySpark, PyTorch, Scikit-Learn, TensorFlow, Hive, Oozie, Hadoop, Python, SQL, Linux

### Data Scientist – Machine Learning

May 2016 – Oct 2017

*Wellcare Health Plans (Centene), Tampa FL*

*On-Site*

- Reduced hospital readmissions by 60% using XGBoost and Scikit-Learn, directly limiting Medicare penalty exposure for the plan.
- Key contributor to a \$1M/year capital project coordinating Informatica, Hadoop, SAS, and Python workflows across external vendors and multiple business units.
- **Stack:** PySpark, Hive, Python, R, SQL, XGBoost, Scikit-Learn

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>Sr. Decision Support Analyst</b><br><i>Gateway HealthPlan (Highmark), Pittsburgh PA</i>                                                                                                                                                                                                                           | <b>Jan 2015 – Apr 2016</b><br><i>On-Site</i> |
| <ul style="list-style-type: none"> <li>Worked directly with the CFO to resolve reconciliation gaps between insurance claims and accounting systems via ETL improvements; built ARIMA cash-flow forecasting model in R; managed a team of three data engineers and supervised SAS platform administration.</li> </ul> |                                              |
| <b>Database &amp; ETL Analyst</b><br><i>UPMC Health Plan, Pittsburgh PA</i>                                                                                                                                                                                                                                          | <b>Aug 2012 – Oct 2014</b><br><i>On-Site</i> |
| <ul style="list-style-type: none"> <li>Built enterprise-wide ETL reporting layers in Oracle SQL and SAS; developed a Random Forest model to segment high-cost members. Earned the 2014 ACES Award (top 1% of UPMC employees) for excellence in service and process improvement.</li> </ul>                           |                                              |
| <b>Project Manager</b><br><i>University of Pittsburgh, Pittsburgh PA</i>                                                                                                                                                                                                                                             | <b>Jan 2011 – Aug 2012</b><br><i>On-Site</i> |
| <ul style="list-style-type: none"> <li>Managed two \$1.5M grant-funded research projects end-to-end (20+ staff), completed data collection ahead of schedule and under budget; projects featured on local television.</li> </ul>                                                                                     |                                              |

## Projects

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>NameGen: Generative AI Name Generator</b>   <i>Python, FastAPI, Docker</i>                                                                                                                                                                                                                                                                                                                                                  | <a href="#"><u>Source Code</u></a> |
| <ul style="list-style-type: none"> <li>Generates novel names using directed-graph letter-transition probabilities sampled via multinomial distribution – the same statistical foundation underlying language model sampling.</li> <li>Deployed three ways: interactive CLI (with pickle-cached graph for speed), local FastAPI service, and Dockerized container, demonstrating a complete ML deployment lifecycle.</li> </ul> |                                    |
| <b>E-Commerce Fraud Detection</b>   <i>Python, XGBoost, Optuna, joblib</i>                                                                                                                                                                                                                                                                                                                                                     | <a href="#"><u>Source Code</u></a> |
| <ul style="list-style-type: none"> <li>Three-model ensemble (Random Forest, Isolation Forest, XGBoost) applied to e-commerce fraud; XGBoost hyperparameters tuned with Optuna.</li> <li>Models serialized with joblib pipelines, ready for production deployment.</li> </ul>                                                                                                                                                   |                                    |

## Education

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|--------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Duquesne University, Pittsburgh PA USA</b><br><i>Masters in Computational Mathematics</i>     | <b>2008 - 2011</b><br><i>3.9 GPA</i> |
| <b>KDI School, Seoul (Relocated to Sejong City), S. Korea</b><br><i>Masters in Public Policy</i> | <b>2004 - 2005</b><br><i>3.9 GPA</i> |
| <b>Universidad de los Andes, Bogota, Colombia</b><br><i>Masters in Economics</i>                 | <b>2001 - 2003</b><br><i>3.7 GPA</i> |
| <b>Universidad de los Andes, Bogota, Colombia</b><br><i>Bachelors in Economics</i>               | <b>1996 - 2001</b><br><i>3.7 GPA</i> |

## Certificates

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|------------------------------------------------|--------------------|
| <b>AWS Certified Data Engineer – Associate</b> | <b>In Progress</b> |
| <b>AWS Certified Cloud Practitioner</b>        | <b>2022</b>        |
| <b>Tensor Flow Specialization</b>              | <b>2020</b>        |
| <b>Deep Learning Specialization</b>            | <b>2018</b>        |
| <b>Machine Learning Specialization</b>         | <b>2017</b>        |
| <b>Machine Learning Certificate</b>            | <b>2015</b>        |

## Technical Skills

**Languages:** Python, SQL, Java, R, SAS, Rust, Matlab  
**Big Data & Pipelines:** Apache Hadoop, Hive, PySpark, Oozie, Impala, Sqoop, Flume  
**Cloud & Databases:** AWS, Oracle DB, MS SQL Server, MySQL  
**Machine Learning:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, Optuna, SAS Viya, PySpark MLlib  
**GenAI & NLP:** LLM APIs, prompt engineering, NLP feature engineering, generative modeling  
**MLOps & Tools:** FastAPI, Docker, Flask, Git, GitHub, GitLab, BitBucket, Postman, Jira

## Selected Publications

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|-----------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Participatory Assessment of the Health of Latino Immigrant Men in a Community</b><br><i>Journal of Immigrant and Minority Health</i> | <b>2013</b> |
| <b>Bayesian Regression Inference Using a Normal Mixture Model</b><br><i>Masters Thesis, Duquesne University</i>                         | <b>2012</b> |
| <b>FARC Terrorism in Colombia: A Clustering Analysis</b><br><i>CEDE Working Papers Series</i>                                           | <b>2004</b> |